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Effects of Indoor Air Pollution and How To Control Them

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Abstract: In the last two decades pollution level in many cities has broken all previous records. The air quality in the city has been dropped it to a level that can cause breathing discomfort to people with lungs, asthma and heart diseases. Assessment of air pollution in Indian cities (Jan., 2017) revealed that 1.2 million persons die every year due to outdoor air pollution in India. As a result of these, where people love to go outside for enjoyment, now they have closed themselves in the walls of concrete. Majority of people spend the bulk of their time indoors whether it's at home or working in an office environment. In today scenario, the buildings are designed on the concept of energy efficient. It has been determined that these air- tight sealing of building leads to various health problems such as itchy eyes, skin rashes, headache, allergy etc. Man himself makes a source of indoor air pollution. Aim is to examine how indoor plants can help to reduce indoor air pollution.

Keywords: *Indoor plants, Air pollution, Indoor air quality.*

I. INTRODUCTION

Indoor air pollution is becoming the biggest environmental problem of the world. There is a myth that air pollution occurs either in outdoor environment or in industrial environment. If you think that you are safe at home and away from outside pollution. The truth is, you are not. Air inside the conditioned space can be substantially more polluted than outdoor air. Every year, 4.3 million people die due to the exposure of household air pollution caused by indoor open air (WHO, 2015).

Indoor air pollution came to notice when more than hundred people fell sick within fifteen minutes of entering the headquarters of EPA building at Washington D.C., due to accumulation of carbon monoxide and poor ventilation. After occurrence of such incidences, concept of IAQ was first introduced among scientific community in 1980.

II. INDOOR AIR QUALITY

It refers to the nature of the conditioned (heat/cool) air that circulates throughout space/area, where we work and live i.e. the air we breathe most of the time.

III. SOURCES OF INDOOR AIR POLLUTION

Indoor pollution may be result of air tightness of buildings, poorly designed, air conditioning and ventilation systems, indoor sources of pollution and outdoor sources of pollution. It can be ten times worse than outdoor air pollution.

A. Air tightness of building

In response to making building energy efficient and conserve energy resources, buildings and homes are designed having tighter construction which seal it from outside air. It causes inadequate supply of fresh air, as a result, negative pressure develops which causes

1. ground level pollutants (CO, Radon etc. to be drawn inside the building),
2. release of odour (Bioaerosols),
3. pull outside polluted air from vents, cracks and openings and increase dust, pollen etc.,
4. may results in "Sick building syndrome".

People who spend 70-90% of their time indoor may face various health problems due to indoor air pollution.

B. Sources of pollution in typical residential building

Sources of emissions that cause indoor air pollution:

	Emission					
Source	Formalde-hyde	Xylen/ toulene	Benzene	Ammonia	Alcohol	Ace-tone
Adhesives	•	•	•		•	
Bioeffluents[1]		•		•	•	•
Carpeting					•	
Caulking compounds	•	•	•		•	
Ceiling tiles	•	•	•		•	
Cleaning products				•		
Cosmetics					•	•
Draperies	•					
Electro photographic printers		•	•	•		
Fabrics	•					
Facial tissue	•					
Floor coverings	•	•	•		•	
Grocery bags	•					
Nail polish remover						•
Office correction fluid						•
Paints	•	•	•		•	
Paper towels	•					
Particle board	•	•	•		•	
Stains and varnishes	•	•	•		•	
Upholstery	•					
Wall coverings		•	•		•	

[1] *substances emitted by people through normal biological processes such as respiration.*

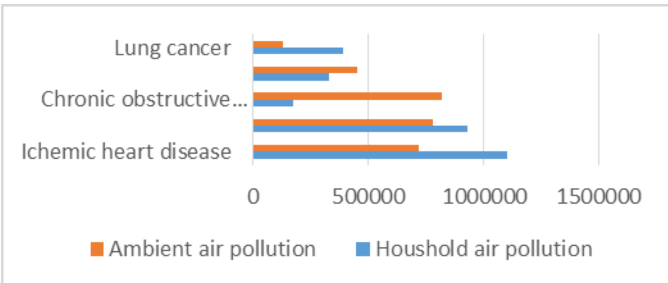
Source: B.C. Wolverton, PhD; How to grow fresh air; Penguin, 1997.

C. Poorly designed air conditoing and ventilation system

It may results into the production of fungi, molds and other sickness causing microbes.

IV. EFFECTS OF INDOOR AIR POLLUTION

WHO has reported in his data of 2015 that more people died in 2013 was due to air pollution which indicates that indoor air pollution is much more dangerous.



Household air pollution caused by burning solid fuels like coal, wood and dung ambient air pollution caused by emissions from power generation, transportation, agriculture etc.

Source: WHO report, 2015

Using of synthetic materials such as formaldehyde, pressed wood products, surface finishes etc. in the construction of furniture or other components of buildings are the primary source of indoor air pollution. People who are exposed to these, can develop serious symptoms including eye, nose, asthma, headache, skin irritation etc.

A recent survey conducted by Artemis Hospitals at Gurgaon as part of Clean Air India Movement (CLAIM) says that about 76 per cent of the offices and houses, 3 had PM level far above 2.5 the normal (>60 µg/m). Due to unhealthy IAQ, about 31 per cent of the respondents had some kind of airway disease and 46 per cent people were found to have symptoms suggesting a respiratory disease (Hindu, 2016). The pollutant released indoors is 1,000 times more likely to affect human lungs than a pollutant released outdoors.

V. Indoor Plant

Bringing a bit of nature indoors is the best way to improve the quality of indoor air. The International Space Station – NASA discovered that plants help reduce VOC's (volatile organic chemicals/ compounds) and can be efficient way to filter air in living compartments.

Plants absorb volatile organic compounds from the air into their leaves and then translocate them to their root zone, where microbes break them down. Microorganisms in the soil can use trace amounts of pollutants as a food source. Some organic chemicals absorbed by plants from the air are destroyed by the plant's own biological processes. The plant's roots take up aqueous solutions in the rooting media. Given that air also reaches plant roots, uptake by root tissues is another means by which air can be purified.

VI. Benefits of Indoor Plant

Today there are numerous scientific studies documenting the benefits of indoor plant (for summaries, see: Lohr, 2000; Pearson-Mims and Lohr, 2000; Relf and Lohr, 2003). The range of benefits that has been documented is broad: air quality is improved (Wood et al., 2002), stress is lowered (Dijkstra et al., 2008), recovery from illness is faster (Ulrich, 1984), mental fatigue is reduced (Tennessen and Cimprich, 1995), and productivity is higher (Lohr et al., 1996) which are in the following ways:

A. *Indoor air quality:*

Plants release oxygen and moisture into the air. Some of these changes to the environment can increase the health and comfort level for humans in enclosed environments.

B. *Air pollutant:*

The studies showed that many common foliage plants reduced levels of some interior pollutants, including formaldehyde and carbon monoxide.

C. *Relative humidity:*

When buildings are being heated, the relative humidity of the inside air is often below the range recommended for human comfort i.e. 30-60%. Presence of plants may increase the relative humidity by 5%, if less than 2% of space is occupied by the plant.

D. *Particulate matter:*

Adding plants to the periphery of a room reduced particulate matter deposition by as much as 20%, even in center of the room many meters from the plants.

E. *Acoustics:*

Plants can reflect, diffract, or absorb sounds, depending on the frequency. Generally, the researchers found that plants worked best at reducing high frequencies sounds in rooms with hard surfaces; the effect was similar to adding carpet.

F. *Well-being:*

Presence of plants in the room makes human more attentive, friendly. It provide aesthetic and make it inviting place to live or work.

G. *Stress reduction:*

Images of nature, interior plants evoke stress reducing effects that are similar to those evoked by nature.

H. *Productivity:*

Productivity has been shown to be higher when plants are present. People may respond significantly more quickly. It helps to improve employee performance and reduce absenteeism.

I. *Health improvement:*

The positive benefits of plants are not simply associated with their decorative value, it can use as a distraction to help keep one's mind off of the discomfort, because colorful objects were not as effective in influencing pain tolerance.

VII. Selection of Indoor Plant

Selection of indoor plants to mitigate air pollution in offices and homes depends not only on their ability to clean the air but also on their growth habit. NASA scientists suggests that the most sophisticated pollution-absorbing device in your home is a potted plant which are in the following ways:

A. *Areca palm(Dypsis lutescens):*

Areca palm is the best plant for removing a variety of toxins (formaldehyde). It likes bright, indirect light. Because of a high transpiration rate, it adds a lot of humidity to the air and needs to be watered regularly.

B. *Snake plant (Sansevieria trifasciata):*

It clears smog, formaldehyde, trichloroethylene from air. It produces oxygen and removes carbon dioxide at night time, making it ideal for bedrooms and other light room.

C. *Spider plant (Chlorophytum comosum):*

It removes smog, formaldehyde, benzene and xylene found in auto exhaust, synthetic perfume and paint. NASA study found that this plant can remove 96% of carbon monoxide within a sealed chamber. It prefers medium to bright light but avoid extended amount of direct sun.

D. *Weeping fig (Ficus benjamina):*

It removes formaldehyde, xylene and fluoene. They enjoy full to semi-sun and moist soil.

E. Broadleaf Lady Palm (Rhapis excelsa):

It helps to filter out ammonia, a chemical found in cleaning products. As they're also good with humidity and thrive in low light, they're best placed in the bathroom.

F. Aloe vera (Aloe barbadensis):

It is easy to grow in indoor sunlight, and needs to be watered only once a week. Aloe helps to keep your home free from benzene, which is commonly found in paint and certain chemical cleaners.

G. Rubber plant (Ficus robusta):

This houseplant releases more oxygen than most other plants, and purifies indoor air by removing chemicals such as formaldehyde. It grow well indoors, preferring moderate light.

H. Dragon plant (Dracaena marginata):

The dragon tree likes indirect sunlight, and need to be watered sparingly. It eliminates formaldehyde, xylene, toluene, benzene, and trichloroethylene.

I. Money plant/ golden pothos (Epipremnum aureum):

It is also considered one of the most effective indoor air purifiers for removing common toxins. It eliminates formaldehyde, xylene, toluene, benzene, carbon monoxide.

J. Boston fern (Nephrolepis exaltata "Bostoniensis"):

It act as act as a natural air humidifier, the Boston Fern removes formaldehyde and is a general air purifier.

CONCLUSION

We came to know that how enclosed spaces inhabited by humans produce following effects such as reduction in oxygen level of spaces, increase in - carbon dioxide, temperature, humidity, odor and bioaerosols.

We can conclude that indoor plants are not just improving your indoor aesthetic but also provide health benefits. Due to lack of awareness, people don't know how to mitigate these indoor air pollution. Living in green and flowering plants have an amazing ability to remove a number of truly awful toxic chemicals from the air. You can use plants in your home or office to improve the quality of the air and to make it a more pleasant place to live in and work, where people feel better, perform better and enjoy life more. In developing countries, plant will serve as a cost-effective tool to mitigate indoor air pollution where expensive pollution mitigation technology may not be economically feasible.

References

[1] Dahiya, S., Myllyvirta, L., & Sivalingam, N. (2017). *Airpocalypse: Assessment of air pollution in Indian cities.* New Delhi: GPET.

[2] Heather L.Papinchak, E. H. (April-June 2009). Effectiveness of houseplants in reducing the indoor air pollutant ozone (Vol. 19).

[3] I.Lohr, V. (2010). What are the benefits of plants indoors and why do we respond positively to them? 675-682. Retrieved from www.actahort.org

[4] (2016). *International Summit.* New Delhi: Indian Pollution Control Association.

[5] Kankaria, A., Nongkynrih, B., & Gupta, S. K. (Oct.-Dec. 2014). Indoor air pollution in India: Implications on health and its control. *Indian Journal Community Medical*, 203-207.

[6] Torpy, F., & Zavattaro, M. (2018). Bench- study of green wall plants for indoor air pollution reduction. *Journal of living architecture, 5*, 1-15.

[7] Wolverion, B., Johnson, A., & Bounds, K. (1989). Interior landscape plants for indoor air pollution abatement. NASA.